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Long-Term Outcomes of a Structured Lifestyle-Based Nutritional Intervention for Obesity Management: A 10-Year Observational Study

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Abstract

Background: Obesity is a chronic, relapsing disease associated with substantial metabolic morbidity. While bariatric surgery and pharmacotherapy can be effective, their use may be limited by invasiveness, adverse effects, cost, or long-term accessibility. There remains a need for sustainable, non-invasive lifestyle-based strategies applicable in routine outpatient care. This study evaluated 10-year outcomes of a structured lifestyle-based nutritional intervention (metabolic endolipolysis) in adults with obesity.

Methods: This retrospective observational cohort study evaluated long-term outcomes of a structured outpatient nutritional intervention in a real-world clinical setting. A total of 100 adults with obesity (BMI ≥ 30 kg/m²) were included. The intervention was based on a defined macronutrient composition with strict dietary fat limitation (approximately 30 g/day), adequate protein intake (≥110 g/day), and normoglycemic carbohydrate intake, combined with behavioral support and recommended daily moderate physical activity. Body composition was assessed annually using bioelectrical impedance analysis. Metabolic outcomes included HbA1c, fasting glucose, lipid profile, and liver enzymes.

Results: A mean fat mass reduction of 9.6 kg was observed during the first year, with sustained reductions over the 10-year follow-up. Total fat mass decreased by 23.1%, including a 32.3% reduction in visceral fat, while lean body mass was largely preserved. Improvements were observed in metabolic parameters, including reductions in HbA1c (−8.5%), LDL cholesterol (−36.6%), and total cholesterol (−30.0%). The greatest changes occurred within the first year and remained relatively stable thereafter. Adherence and retention appeared relatively high based on attendance at scheduled visits and self-monitoring; however, quantitative adherence metrics and validated assessment tools were not available.

Conclusions: This long-term observational study suggests that a structured lifestyle-based nutritional intervention delivered in an outpatient setting may represent a feasible, non-invasive component of obesity management. The intervention was associated with sustained improvements in body composition and selected metabolic parameters. These findings should be interpreted within the limitations of an observational design, including the absence of a control group, and warrant confirmation in randomized controlled trials.

Trial registration: Retrospective observational study; registration not required.

Keywords: Obesity; Lifestyle intervention; Nutritional intervention; Dietary macronutrient composition; Dietary fat restriction; Body composition; Visceral adiposity; Long-term outcomes.

Introduction

Obesity is a chronic, relapsing disease associated with substantial metabolic morbidity and is increasingly recognized as an adiposity-based chronic disease (ABCD). The World Health Organization estimates that more than 650 million adults worldwide are affected, with prevalence continuing to rise globally. This growing burden is associated with significant healthcare costs and is strongly linked to type 2 diabetes, cardiovascular disease, and premature mortality.

Current therapeutic options for obesity include lifestyle interventions, pharmacotherapy, and bariatric surgery. Although pharmacological agents, such as glucagon-like peptide-1 (GLP-1) receptor agonists, have demonstrated substantial efficacy, their long-term use may be limited by accessibility, tolerability, and cost. Bariatric (metabolic) surgery is highly effective in inducing sustained weight loss and provides metabolic benefits beyond caloric restriction; however, it carries procedural risks and may not be accessible, acceptable, or appropriate for all patients. These considerations highlight the need for effective, sustainable, and non-invasive approaches that can be implemented in routine clinical practice.

Metabolic endolipolysis was developed as a structured outpatient nutritional intervention based on a defined macronutrient composition with strict dietary fat limitation, combined with behavioral support and adherence monitoring. The term “metabolic endolipolysis” is used here as an operational designation of this nutritional approach rather than a formally recognized clinical entity.

The present study evaluates the long-term outcomes of this structured lifestyle-based nutritional intervention in a real-world clinical setting over a 10-year follow-up period. The aim of the study was to assess its association with sustained changes in body composition and selected metabolic parameters, as well as its feasibility within routine outpatient care.

An overview of the study design is presented in Fig. 1.

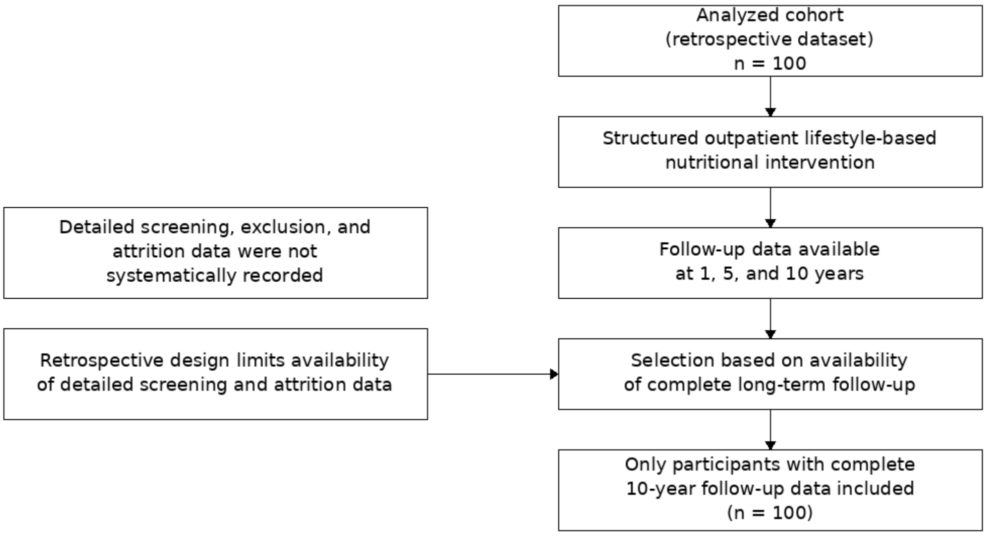

Fig. 1. Flow diagram of participant inclusion and follow-up.

The diagram illustrates the retrospective cohort structure and the availability of longitudinal data over the 10-year follow-up period. Due to the retrospective design, complete data on initial screening, exclusions, and attrition were not available, which may have introduced selection bias. Only participants with complete 10-year follow-up data were included in the analysis, potentially resulting in overrepresentation of participants with complete long-term follow-up, which may have contributed to selection bias and overestimation of long-term effectiveness.

Methods

Study design and participants

This retrospective observational study included 100 participants (aged 25–65 years) enrolled in a structured outpatient nutritional program (metabolic endolipolysis).

Inclusion criteria comprised adults with obesity (BMI ≥ 30 kg/m²). Exclusion criteria included previous bariatric surgery, severe comorbidities requiring surgical treatment, endocrine disorders (e.g., thyroid or adrenal dysfunction), pregnancy or lactation, use of medications known to affect body weight, and acute or chronic conditions potentially influencing metabolic outcomes. Inability to adhere to dietary or behavioral recommendations was also considered an exclusion criterion.

Due to the retrospective design, detailed screening and exclusion data were not systematically recorded and could not be reliably reconstructed. Consequently, the initial number of individuals assessed for eligibility and attrition rates were unavailable, and only patients with complete 10-year follow-up were included in the present analysis. The cohort thus represents participants with complete long-term follow-up data. This may have resulted in overrepresentation of highly adherent individuals and may have contributed to overestimation of long-term effectiveness.

Longitudinal clinical and metabolic data were available for all included participants, who were followed for 10 years. Baseline characteristics of the study population, including age, sex distribution, and selected metabolic parameters, are summarized in Supplementary Table S0.

The nutritional program was delivered as part of routine outpatient clinical care and was not implemented for research purposes or as a protocol-driven intervention. No prospective allocation, randomization, or investigator-directed intervention was performed, and all clinical decisions were made as part of standard patient care. The present analysis represents a retrospective evaluation of routinely collected clinical data without modification of treatment for research purposes.

Dietary intervention

Participants followed a structured outpatient nutritional program based on the principles of metabolic endolipolysis, defined as a low-fat, normoglycemic, non-ketogenic dietary regimen combined with continuous behavioral support. Behavioral strategies included regular counseling sessions, self-monitoring, and motivational reinforcement aimed at supporting long-term adherence. Dietary intake was monitored to support adherence to the prescribed macronutrient composition and energy targets.

Adherence was assessed based on attendance at scheduled visits and self-monitoring records. However, no validated adherence instruments were used, and quantitative adherence metrics (e.g., proportion of attended visits or compliance with dietary and physical activity targets) were not systematically recorded. Therefore, adherence could not be quantitatively assessed and should be interpreted with caution.

The nutritional intervention emphasized a defined macronutrient composition designed to support gradual fat loss while preserving lean body mass. Protein intake was prescribed at a minimum target of 110 g/day, with minor day-to-day variation permitted within the predefined macronutrient framework.

Average daily energy intake was approximately 2,010 kcal, calculated from digestible macronutrients (110 g protein, 320 g carbohydrates, and 30 g fat). Energy intake was individualized based on estimated total energy expenditure (TEE), aiming to achieve a moderate caloric deficit, particularly during the initial phase of the intervention. However, TEE was not systematically quantified, and the magnitude of the energy deficit cannot be precisely determined. Intake was subsequently adjusted according to clinical response, although these adjustments were not systematically quantified.

Dietary fiber intake ranged from 27 to 38 g/day and was not included in total energy calculations. Hydration was maintained at approximately 30 mL/kg of ideal body weight.

Physical activity and behavioral support

Participants were encouraged to engage in moderate daily physical activity (≥45 minutes/day), including walking, swimming, or cycling. Behavioral support was provided through regular consultations conducted by trained healthcare professionals (physicians or nutrition specialists), typically at intervals of 1–3 months depending on individual needs. These consultations focused on adherence, motivation, and sustainable lifestyle modification. No pharmacological or surgical weight-loss interventions were prescribed as part of routine clinical care.

Measurements and outcomes

Body composition was assessed annually using bioelectrical impedance analysis (BIA; Tanita MC-980, Tokyo, Japan), including total body fat, visceral fat, and lean body mass. Raw BIA parameters (e.g., resistance, reactance, phase angle) were not consistently available across the full follow-up period and are therefore not reported.

Anthropometric measurements were obtained at baseline, 6 months, 12 months, and annually thereafter.

Fasting blood samples were collected to assess glucose, HbA1c, HOMA-IR, total cholesterol, LDL cholesterol, triglycerides, and liver enzymes (ALT, AST, GGT). All biochemical analyses were performed in an accredited laboratory using standardized procedures.

Primary outcomes included changes in total and visceral fat mass, glycemic control, and lipid profile. Secondary outcomes included preservation of lean body mass and changes in liver enzymes.

Statistical analysis

Statistical analyses were performed using IBM SPSS Statistics version 27.0 (IBM Corp., Armonk, NY, USA). Continuous variables are presented as mean ± standard deviation (SD). Changes over time were evaluated using repeated-measures analysis of variance (ANOVA). Assumptions of normality and sphericity were assessed where appropriate. Statistical significance was set at $p < 0.05$. Missing data (<5%) were handled using multiple imputation with predictive mean matching.

A formal a priori sample size calculation was not performed due to the retrospective nature of the study and should be considered a methodological limitation when interpreting the findings. The sample size was therefore determined by the availability of complete long-term follow-up data.

Results

Body Composition

Participants exhibited sustained reductions in body weight and adiposity over the 10-year follow-up, while lean body mass remained largely stable (Table 1). Mean body weight decreased from 98.4 ± 12.3 kg at baseline to 83.7 ± 10.9 kg at year 10, corresponding to a relative reduction of 14.9% ($p < 0.001$).

Fat mass decreased by 9.6 kg during the first year and by a total of 23.1% over the study period ($p < 0.001$). Visceral fat, assessed by bioelectrical impedance analysis, declined progressively, reaching a total reduction of 32.3% at year 10 relative to baseline ($p < 0.001$).

Lean body mass did not change significantly throughout the follow-up period. However, the lack of detailed early-phase data prevents assessment of potential transient lean mass loss during the initial phase of weight reduction.

Changes in anthropometric and metabolic parameters at years 1, 5, and 10 are summarized in Table 1, and relative percentage changes over time are presented in Table 2.

Table 1. Changes in anthropometric and metabolic parameters over 10-year follow-up

Parameter	**Baseline (mean ± SD)**	**Year 1**	**Year 5**	**Year 10**	**p-value (vs baseline)**
Body weight (kg)	98.4 ± 12.3	90.9 ± 11.9	90.5 ± 11.4	83.7 ± 10.9	< 0.001
BMI (kg/m²)	36.1 ± 3.8	33.2 ± 3.5	31.5 ± 3.4	30.2 ± 3.4	< 0.001
Fat mass (kg)	41.6 ± 7.8	37.0 ± 7.2	34.2 ± 6.5	32.0 ± 6.0	< 0.001
Visceral fat (cm², BIA)	155 ± 22	132 ± 20	120 ± 18	105 ± 16	< 0.001
HbA1c (%)	5.9 ± 0.4	5.6 ± 0.3	5.6 ± 0.3	5.4 ± 0.3	0.009
LDL	4.1 ± 0.6	3.3 ± 0.5	3.0 ± 0.5	2.6 ± 0.4	< 0.001

cholesterol (mmol/L)					
Total cholesterol (mmol/L)	6.0 ± 0.7	4.9 ± 0.6	4.7 ± 0.6	4.2 ± 0.5	< 0.001

Values are presented as mean ± SD. Statistical significance was assessed using repeated-measures ANOVA with comparisons to baseline; Greenhouse–Geisser correction was applied where appropriate.

Abbreviations: BMI, body mass index; HbA1c, glycated hemoglobin; BIA, bioelectrical impedance analysis; SD, standard deviation.

Table 2. Percentage changes in anthropometric and biochemical parameters over 10 years

Parameter	Year 1 (%)	Year 5 (%)	Year 10 (%)	p-value (vs baseline)*
Body weight	−7.7	−8.0	−15.0	< 0.001
BMI	−8.0	−12.8	−16.3	< 0.001
Fat mass	−11.0	−17.9	−23.1	< 0.001
Visceral fat (BIA)	−14.8	−22.6	−32.3	< 0.001
HbA1c	−5.1	−5.1	−8.5	0.009
LDL cholesterol	−19.5	−26.8	−36.6	< 0.001
Total cholesterol	−18.3	−21.7	−30.0	< 0.001

Percentage changes were calculated relative to baseline values and are presented for descriptive purposes only. *p-values correspond to comparisons with baseline based on absolute values (see Table 1 for absolute values).

Abbreviations: BIA, bioelectrical impedance analysis; HbA1c, glycated hemoglobin.

Metabolic Parameters

Metabolic parameters improved over time (Table 1). HbA1c decreased from 5.9 ± 0.4% at baseline to 5.4 ± 0.3% at year 10 (−8.5%, $p = 0.009$). LDL cholesterol declined from 4.1 ± 0.6 mmol/L to 2.6 ± 0.4 mmol/L (−36.6%, $p < 0.001$), and total cholesterol decreased from 6.0 ± 0.7 mmol/L to 4.2 ± 0.5 mmol/L (−30.0%, $p < 0.001$). Liver enzyme levels (ALT, AST, GGT) remained within reference ranges throughout the study period.

Yearly trajectories of anthropometric and metabolic parameters are provided in Supplementary Table S3.

Adherence and Follow-up

Adherence and retention appeared to be relatively high based on attendance at scheduled visits and completion of self-monitoring records; however, no validated adherence instruments or quantitative adherence metrics were available. These findings should therefore be interpreted with caution, particularly given that the analysis included only participants with complete long-term follow-up.

Discussion

The present study provides long-term observational evidence that a structured lifestyle-based nutritional intervention delivered in an outpatient setting was associated with reductions in fat mass. Sustained decreases in body weight and adiposity were observed over a 10-year follow-up, while lean body mass was largely preserved. However, the lack of detailed early-phase data prevents assessment of potential transient lean mass loss during the initial phase of weight reduction.

Taken together, these findings suggest that this approach may represent a feasible component of long-term obesity management in selected clinical settings and could complement existing therapeutic strategies in selected patient populations [1,11]. However, causal relationships cannot be established due to the observational study design.

Importantly, although significant reductions in adiposity were observed, the mean BMI remained within the obese range at the end of follow-up. Despite sustained improvements, the persistence of obesity at the group level indicates that this intervention should be interpreted as a strategy for long-term disease management rather than disease resolution. This reflects the chronic and relapsing nature of obesity and suggests that the intervention was associated with partial improvement rather than complete remission.

The lifestyle-based intervention described in this study was delivered in an outpatient setting and was based on strict dietary fat limitation, a physiologically defined macronutrient composition, and continuous behavioral support. These characteristics differentiate it from surgical approaches in terms of mechanism and clinical trajectory.

The observed long-term changes in adiposity and selected metabolic outcomes are broadly consistent with findings from other lifestyle-based interventions, although differences in study design, population characteristics, and intervention intensity should be considered [5,6]. While bariatric procedures typically induce greater initial weight loss, structured lifestyle interventions may represent a non-invasive option for selected patients and may contribute to longer-term metabolic stabilization.

The mechanisms potentially underlying the observed associations may relate to the specific macronutrient composition, characterized by reduced dietary fat intake combined with adequate carbohydrate and protein availability [3,7,8]. Although this macronutrient composition differs from general dietary recommendations, it was applied within a structured and supervised clinical setting, which may have supported adherence, safety, and long-term feasibility. However, these mechanistic considerations remain speculative and cannot be confirmed within the present observational study design. The observed improvements in HbA1c and lipid outcomes are consistent with previously reported effects of lifestyle-based interventions on cardiometabolic risk [6,8].

Importantly, these findings should be interpreted in the context of individualized energy intake designed to achieve a caloric deficit, rather than as a consequence of macronutrient composition alone. The relatively high reported energy intake should be viewed in relation to individualized energy requirements and potential reporting limitations inherent to retrospective dietary data.

From a clinical perspective, adherence and retention appeared to be relatively high based on attendance at scheduled visits and self-monitoring. However, these measures were not assessed using validated or quantitative instruments and should therefore be interpreted with caution. Observational indicators, such as long-term follow-up completion and regular attendance at scheduled visits, may suggest sustained engagement among participants included in the analysis.

In the broader therapeutic context, these findings support a potential role of non-surgical and non-pharmacological approaches as part of comprehensive obesity management. While pharmacological treatments such as GLP-1 receptor agonists demonstrate substantial efficacy, their long-term use may be limited by accessibility, tolerability, and cost. Structured lifestyle-based nutritional interventions may therefore represent a pragmatic component of obesity care in routine clinical practice.

The relatively homogeneous study population may limit the generalizability of the findings. In addition, unmeasured confounding factors, including potential changes in concomitant treatments, cannot be excluded.

In summary, this long-term observational study suggests that a structured lifestyle-based nutritional intervention delivered in an outpatient setting may represent a feasible and potentially sustainable component of obesity management [5,6,8]. These findings should be interpreted cautiously given the observational design and the absence of a control group and should be considered hypothesis-generating pending confirmation in prospective controlled studies [3,6,8,11].

Clinical implications

Structured lifestyle-based nutritional interventions delivered in an outpatient setting may represent a pragmatic component of long-term obesity management for selected patients, particularly those who are unwilling or ineligible to undergo surgical treatment or who prefer non-invasive approaches [5,6,8]. Integration of such programs into primary and preventive care settings may help improve access to obesity management and address the long-term burden of metabolic disease.

By combining a defined macronutrient composition within a non-ketogenic dietary model and continuous behavioral support, this type of intervention may complement established surgical and pharmacological therapies within a comprehensive obesity care framework. Such approaches may be particularly relevant for individuals with contraindications to surgery or for those requiring sustained lifestyle support over time.

Strengths and limitations

A major strength of this retrospective observational study is the exceptionally long follow-up period of ten years, providing valuable insight into the durability of metabolic and anthropometric changes achieved through a structured lifestyle-based intervention in a real-world clinical setting. The assessment of multiple metabolic, biochemical, and body composition parameters allows for a comprehensive evaluation of long-term outcomes. In addition, the outpatient design reflects routine clinical practice and supports the feasibility of implementing such an approach over an extended period.

Several limitations should be acknowledged. The retrospective observational design precludes causal inference, and the absence of a randomized control group limits direct comparability with alternative interventions. In addition, no formal a priori sample size calculation was performed due to the retrospective nature of the study and should be considered a methodological limitation when interpreting the findings.

The study population may be subject to selection bias, as participants may have been more motivated and adherent than the general population. Furthermore, the inclusion of only patients with complete long-term follow-up may have resulted in overrepresentation of highly adherent individuals and potential overestimation of long-term effectiveness. The initial number of screened individuals and attrition rates could not be reconstructed due to the retrospective design, further limiting the evaluation of attrition and the interpretation of cohort representativeness.

Adherence was assessed based on attendance at scheduled visits and self-monitoring records; however, no validated adherence instruments were used, and adherence could not be quantitatively assessed, limiting the interpretability of adherence-related findings. Changes in concomitant pharmacological treatments during follow-up were not systematically recorded and may have influenced the observed metabolic outcomes.

Although additional metabolic and anthropometric parameters (e.g., HDL cholesterol, triglycerides, and waist circumference) were routinely measured in clinical practice, these variables were not consistently available across the full follow-up period and were therefore not included in the present analysis.

Body composition was assessed using bioelectrical impedance analysis, which provides less precise estimates than imaging-based methods such as MRI or dual-energy X-ray absorptiometry (DEXA) [11]. This represents an important methodological limitation, particularly in the context of long-term body composition assessment. Detailed temporal dynamics of lean mass, particularly during early phases of the intervention, were not available, which further limits interpretation. Raw BIA parameters (resistance, reactance, phase angle) were not consistently available across the full follow-up period and could therefore not be included in the analysis, which limits the interpretability of body composition findings.

Despite these limitations, the consistency of findings across anthropometric and metabolic outcomes over a prolonged follow-up period supports the potential clinical relevance of the observed findings. These results should be interpreted within the context of an observational design and considered hypothesis-generating, warranting confirmation in prospective controlled studies.

Conclusion

This long-term observational study suggests that a structured lifestyle-based nutritional intervention delivered in an outpatient setting was associated with sustained reductions in body weight and fat mass over a 10-year follow-up, while lean body mass and overall metabolic stability were largely preserved. Improvements in selected metabolic outcomes were also observed.

Despite these changes, mean BMI remained within the obese range at the end of follow-up, indicating that the intervention was associated with partial improvement rather than disease resolution. These findings should therefore be interpreted within the context of long-term disease management rather than remission.

Given the observational design and absence of a control group, causal relationships cannot be established. The findings should be interpreted with caution and considered hypothesis-generating, requiring confirmation in prospective controlled studies.

Clinical and Public Health Implications

Structured lifestyle-based nutritional interventions delivered under clinical supervision may represent a feasible component of routine outpatient care for selected patients, particularly those who are unwilling or ineligible to undergo surgical treatment or who require long-term metabolic management.

Within a comprehensive obesity care framework, such approaches may complement established surgical and pharmacological strategies rather than replace them. Integration of structured programs into routine clinical practice may help expand access to long-term obesity management, particularly when combined with ongoing behavioral support.

Declarations

Ethics approval and consent to participate

In accordance with applicable Czech legislation and institutional policies governing retrospective analyses of anonymized clinical data at University Hospital Brno, studies based on fully anonymized data do not require formal Ethics Committee approval or individual informed consent. Accordingly, the present study did not require formal ethics approval or informed consent.

The nutritional program was delivered as part of routine outpatient clinical care and was not implemented for research purposes or as a protocol-driven intervention. No prospective allocation, randomization, or investigator-directed intervention was performed, and all clinical decisions were made as part of standard patient care. The study is based exclusively on retrospectively collected data obtained during routine clinical management, and no additional procedures or interventions were introduced for research purposes.

All data were anonymized prior to analysis. The study was conducted in accordance with the principles of the Declaration of Helsinki.

Competing interests

The authors declare that they have no competing interests.

Consent for publication

All authors consent to the publication of this manuscript. Individual patient consent for publication was not required, as only anonymized data were used.

Availability of data and materials

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Authors' contributions

MN and DS conceived the study. MC and LB collected and analyzed the data. JT and LB evaluated laboratory parameters. MN drafted the manuscript. All authors read and approved the final manuscript.

Acknowledgements
None.

Abbreviations

BIA, Bioelectrical Impedance Analysis
BMI, Body Mass Index
HbA1c, Glycated Hemoglobin
HOMA-IR, Homeostasis Model Assessment for Insulin Resistance

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Supplementary Table S0. Baseline characteristics of the study population (n = 100)

Parameter	**Value**
Age (years)	49.8 ± 11.1 (25–65)
Sex (male/female)	45/55
Body weight (kg)	98.4 ± 12.3
BMI (kg/m²)	36.1 ± 3.8
Fat mass (kg)	41.6 ± 7.8
Visceral fat area (cm²)	155 ± 22
HbA1c (%)	5.9 ± 0.4
LDL cholesterol (mmol/L)	4.1 ± 0.6
Total cholesterol (mmol/L)	6.0 ± 0.7

Values are presented as mean ± standard deviation (SD).
Abbreviations: BMI, body mass index; HbA1c, glycated hemoglobin; SD, standard deviation.

Supplementary Table S1a. Reference daily dietary composition of the metabolic endolipolysis program
Prespecified macronutrient composition and hydration parameters used throughout the intervention.

Parameter	**Target intake**
Energy (kcal/day)*	≈2,010
Protein (g/day)	110 (≈22% of total energy)
Carbohydrates (g/day)	320 (≈64% of total energy)
Fat (g/day)	30 (≈13% of total energy)
Fiber (g/day)	27–38
Hydration (mL/kg ideal body weight)	~30

* Total energy intake was calculated exclusively from digestible macronutrients; dietary fiber was not included in total energy calculations. Values represent prespecified reference intakes; minor day-to-day variation was permitted, provided that overall daily energy intake and macronutrient distribution remained within the predefined protocol.

Supplementary Table S1b. Macronutrient-derived energy contribution

Macronutrient	Intake	Energy contribution (kcal)
Protein	110 g	440
Carbohydrates	320 g	1,280
Fat	30 g	270
Total	—	≈1,990

Hydration	~30 ml/kg ideal body weight

Values reflect prespecified reference ranges; minor day-to-day variation was permitted.

Supplementary Table S2. An example of a representative daily meal plan illustrating the dietary framework of the metabolic endolipolysis program

An example of a representative daily meal plan illustrating macronutrient distribution and energy balance within the metabolic endolipolysis program. Macronutrient reference ranges were defined in Supplementary Tables S1 and S1b.

Meal	Food examples	Energy (kcal)	Protein (g)	Carbohydrates (g)	Fat (g)	Notes
Breakfast	Skim/low-fat yogurt (250 g), oats (60 g), banana (150 g)	470	20	78	6	Balanced carb-to-protein ratio
Mid-morning snack	Whole-grain bread (80 g) with cottage cheese 4% (100 g)	260	16	40	6	Support glycemic stability
Lunch	Skinless chicken breast (140 g), cooked rice (200 g dry), vegetables (200 g)	520	34	78	4	Main protein meal
Afternoon snack	Apple (250 g), whey protein drink (15 g protein)	190	17	35	2	Supports satiety
Dinner	White fish (170 g), boiled potatoes (350 g), vegetable salad (150 g)	450	23	80	4	Low-fat, high-protein meal
Evening snack (optional)	Skim milk (300 ml), oat flakes (25 g)	210	10	25	8	Adherence support
Daily total	—	≈2,100	≈120	≈336	≈30	Fiber: 27–38 g/day, hydration: ~30 ml/kg IBW

Total daily energy intake was calculated from digestible macronutrients; dietary fiber was not included in energy calculations. Representative meal plans are provided for illustrative purposes only; individual meals and portion sizes varied, while overall daily macronutrient reference ranges and total energy intake were maintained according to the predefined protocol.

Supplementary Table S3. Yearly changes in anthropometric and metabolic parameters over the 10-year follow-up.

Longitudinal changes in selected anthropometric and metabolic parameters during the 10-year follow-up period.

Year	Body weight (kg, mean ± SD)	Fat mass (kg)	HbA1c (%)	LDL cholesterol (mmol/L)	Total cholesterol (mmol/L)
Baseline	98.4 ± 12.3	41.6 ± 7.8	5.9 ± 0.4	4.1 ± 0.6	6.0 ± 0.7
Year 1	90.9 ± 11.9	37.0 ± 7.2	5.6 ± 0.3	3.3 ± 0.5	4.9 ± 0.6
Year 5	90.5 ± 11.4	34.2 ± 6.5	5.6 ± 0.3	3.0 ± 0.5	4.7 ± 0.6
Year 10	83.7 ± 10.9	32.0 ± 6.0	5.4 ± 0.3	2.6 ± 0.4	4.2 ± 0.5

Values are presented as mean ± SD. Statistical comparisons are reported in the main text (Table 1). Most improvements occurred during the first year and were maintained throughout follow-up.